



Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: <http://www.clinicalnutritionespen.com>

Short Communication

Acute caffeine administration impaired spatial working memory in habitual coffee/tea drinkers: A randomized, placebo-controlled, double-blind study

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ARTICLE INFO

Article history:

Received 12 May 2024

Accepted 27 August 2024

Keywords:

Caffeine

Habitual coffee drinkers

Working memory

Spatial working memory

Spatial span

SUMMARY

Background: Although coffee consumption is widespread worldwide, a recent study showed that acute intake of caffeine negatively affects working memory (WM) performance on n-back tasks among habitual caffeine consumers. However, there is a scarcity of double-blind, placebo-controlled studies that assess the spatial WM (SWM) effects of caffeine using spatial span tasks. Therefore, the present study aimed to examine the effects of acute caffeine administration (200 mg, PO) on SWM and verbal WM (VWM) among habitual caffeine consumers.

Methods: The effects of caffeine on working memory (WM) was evaluated through the administration of backward digit span and spatial span tasks under a delay-dependent condition (0, 4, 8, and 6 s) in a randomized, double-blind, placebo-controlled study involving 18 healthy participants. This data is derived from our previous published study. The total scores obtained and the maximum scores achieved were the primary outcome variables of the study.

Results: Caffeine had a significant impact on SWM (maximum obtained, $p = 0.013$; for total scored, $p = 0.007$) in a delay-independent manner. However, there were no significant main effects of caffeine on VWM ($p = 0.82$ for maximum obtained, $p = 0.56$ for total scored).

Conclusion: Overall, the present findings contradict the commonly held belief that caffeine improves cognitive performance and suggest that acute administration of caffeine may impair SWM in habitual coffee/tea drinkers.

Clinical trial registration number: CT-2018-CTN-02561 (Therapeutic Goods Administration Clinical Trial Registry) and ACTRN12618001292268 (The Australian New Zealand Clinical Trials Registry).

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1. Introduction

Caffeine is the most widely used psychostimulant globally. It is also utilized as a flavoring agent and an additive component in various pharmaceutical formulations in contemporary medicine ([1], see also [2]). The average daily intake of caffeine per individual is estimated to be 70–76 mg, although this can range from 200 to 400 mg depending on the source, location, and cultural context of coffee consumption [3,4]. Likewise, although under-reporting is a concern, the average daily consumption of caffeine by Australian adults was estimated to be from 100 to 180 mg [5] to 200 mg [3].

People often consume caffeine-containing beverages to maintain alertness and enhance cognitive performance. Studies have indicated that low to moderate doses of caffeine (up to 100 mg and 250 mg) can increase alertness [6–10], energetic feelings [11], attention, and vigilance [12,13]. However, the precise effects of caffeine on a wide range of higher cognitive domains remain undetermined (see [2] for review). The challenge in studying the effects of caffeine is that the acute and chronic cognitive effects of prolonged intake of caffeine-containing beverages are not well-established not well-established [14,15].

One approach to evaluate the acute cognitive consequences of caffeine administration in habitual coffee drinkers and predict the effects of long-term exposure is through a randomized clinical trial design. A recent double-blind placebo-controlled study involving

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20 habitual caffeine consumers (about an average of 475 mg/day) found that daily acute caffeine administration (150 mg $\times$ 3 times/day) for 10 days deteriorated working memory (WM) performance on the verbal n-back [16]. Our previous study, which compared the effects of caffeine and dexamphetamine on verbal WM (VWM) and spatial WM (SWM) using forward task conditions of digit (DST) and spatial span (SST), respectively, failed to observe any effects of caffeine [17]. Furthermore, although the study design and tasks applied varied from other studies, another study that used a large sample size did not find any effects of caffeine on VWM and SWM [18]. However, to the best of our knowledge, previous studies that assessed the WM effects of caffeine did not use backward conditions of SST to measure SWM. Therefore, in the present preliminary study, we aimed to investigate and compare the effects of 200 mg caffeine on SWM and VWM by using backward conditions of SST and DST at different delays.

2. Material and methods

Of note, this data is derived from our previous partly published study: a total of twenty participants were recruited for our previous caffeine study comparing the effects of caffeine and d-amphetamine on WM using forward span tasks [17]. The same individuals completed an extra WM task (backward SST and DST), during the same caffeine trial. However, the results of this task have not been shared publicly, as the primary focus of our study was not on habitual caffeine consumption. Now, WM data during SST and DST backward tasks for the habitual caffeine consumers were retrospectively analysed. Subsequently, a total of 18 (5 females) with a mean $\pm$ SD age of 23.1 ± 5.4 years were found to be regular caffeine consumers [1.6 (± 1.1) cups of coffee or tea per day: four of them were using tea, one of them was using either tea or coffee, eleven of them were using coffee, one of them was using either coke or coffee, and one of them was using either coke or tea]. The inclusion and exclusion criteria were clearly outlined before obtaining consent (details regarding the inclusion and exclusion criteria, drug design, and some of the methods can be found in our previous article [17]). The study was approved by The University of Western Australia Human Research Ethics Committee (RA/4/20/4558) and Therapeutic Goods Administration Clinical Trial Registry (CT-2018-CTN-02561).

The study was conducted as a randomized, double-blind, counter-balanced, placebo-controlled cross-over trial with permuted block randomization of drug order. The participants in the study received either placebo or 200 mg of caffeine (PROLAB Nutrition LLC, Chatsworth, USA) orally at 9:30 am during the morning session. The choice of a 200 mg (i.e., relatively low dose) of caffeine was based on previous research studies [6,7,11,12,19,20]: and average daily consumption in some countries, including Australia (for review, see [2,3]).

Caffeine was administered in the form of gelatin capsules (size 0). Additionally, an identical capsule containing powdered glucose served as a placebo. The participants were divided into two groups: half received a placebo on the first day and caffeine on the second day, with a minimum of one week between the two administrations, while the remaining participants received caffeine first and a placebo second. The principal investigator took the role of creating the random allocation sequence and assigning participants to their respective groups. The principal investigator was the only individual who had access to the information that could be used to identify participants during or after data collection. The researchers who were involved in participant recruitment and data collection were kept unaware of the randomization list to ensure the integrity of the trial.

The SST and DST tasks, derived from the Wechsler Adult Intelligence III (WAIS-III) [21], were employed to evaluate the impact of caffeine (200 mg, PO) on WM. As described in our previous report and detailed WM procedures [17], we have applied four delay conditions (0, 4, 6, and 8 s), but using backward DST and SST tasks, to measure WM. Unlike the forward task conditions of span tasks that is thought to measure maintenance component of WM, the backward task conditions, which require further manipulations of the information, measures “manipulation” [22] or “executive functioning” component of WM performance [23,24].

SST was conducted at the 60 min after drug administration. SST is a visuospatial analog of DST. The participants were instructed to touch the same blocks in reverse order after 0, 4, 6, and 8 s of the examiner. During the DST, the digits were articulated at 1-s intervals by the experimenter, and the subject was instructed to repeat the digits in reverse order. This test was administered 120 min after drug administration. The task administration time range was preferred based on the evidence that caffeine, when taken orally, is rapidly absorbed from gastrointestinal tract and reaches its peak plasma concentration in about 30–120 min [25–29].

There were a total of eight items, with the minimum item consisting of two digits and the maximum item comprising nine digits, for each DST and SST. Each item featured two trials of varying lengths but identical digit sequence, resulting in a total of 16 unique trials. The same scoring criteria, consisting of the total scored and maximum obtained, were employed for the two tasks. The total scored refers to the sum of the correct trials from 16 trials until the participant failed the same item on both occasions. The maximum obtained signifies the highest number of trials that the participant accomplished without any errors.

Statistical analysis: The analysis was conducted utilizing R programming version 3.5., and a repeated-measures analysis (ANOVA) and t-test (with Bonferroni correction) were conducted to analyze the data. A Linear mixed-effect model was used to analyse the effects of session (i.e., day order) on WM. Exceptions to these methods are listed in the results.

3. Results

There were significant main effects of caffeine on the maximum obtained ($F [1,17] = 7.7, p = 0.013, \eta^2 = 0.032$; mean $\pm$ SEM of caffeine = 7.68 ± 0.43 , mean $\pm$ SEM of placebo = 8.40 ± 0.31) and total scored ($F [1,17] = 9.5, p = 0.007, \eta^2 = 0.035$; mean $\pm$ SEM of caffeine = 8.30 ± 0.33 , mean $\pm$ SEM of placebo = 8.90 ± 0.27) for SWM: See Fig. 1 for Mean $\pm$ SEM at each delay point for each SWM and VWM. There were also significant effects of delay on the maximum obtained ($F[2.3, 39] = 21.8, p = 1.4 \times 10^{-07}, \eta^2 = 0.25$) and total scored ($F[2.4, 40] = 17.6, p = 1.1 \times 10^{-06}, \eta^2 = 0.19$). However, there were no significant drug by delay interactions for maximum obtained ($F [3,51] = 1.7, p = 0.18, \eta^2 = 0.02$) and total scored $F [3,51] = 1.9, p = 0.141, \eta^2 = 0.01$). The linear mixed-effects showed no effects of session on the total scored ($F[1, 124] = 0.55, p = 0.39, d = 0.17$) and maximum obtained ($F[1, 124] = 0.17, p = 0.67, d = 0.39$). As shown in Fig. 1, pairwise comparisons using paired t-tests (with Bonferroni corrections) showed that there were significant effects of caffeine on maximum obtained ($p = 0.01$) and total scored ($p = 0.006$) at 6 s delay condition. Unlike SWM, there were no significant main effects of caffeine on the maximum obtained ($F [1,17] = 0.05, p = 0.82, \eta^2 = 0.0002$; mean $\pm$ SEM of caffeine = 7.22 ± 0.49 , mean $\pm$ SEM of placebo = 7.14 ± 0.45) and total scored ($F [1,17] = 0.34, p = 0.56, \eta^2 = 0.002$; mean $\pm$ SEM of caffeine = 8.15 ± 0.46 , mean $\pm$ SEM of placebo = 7.94 ± 0.41) for VWM. There were significant effects of delay on the total scored (F

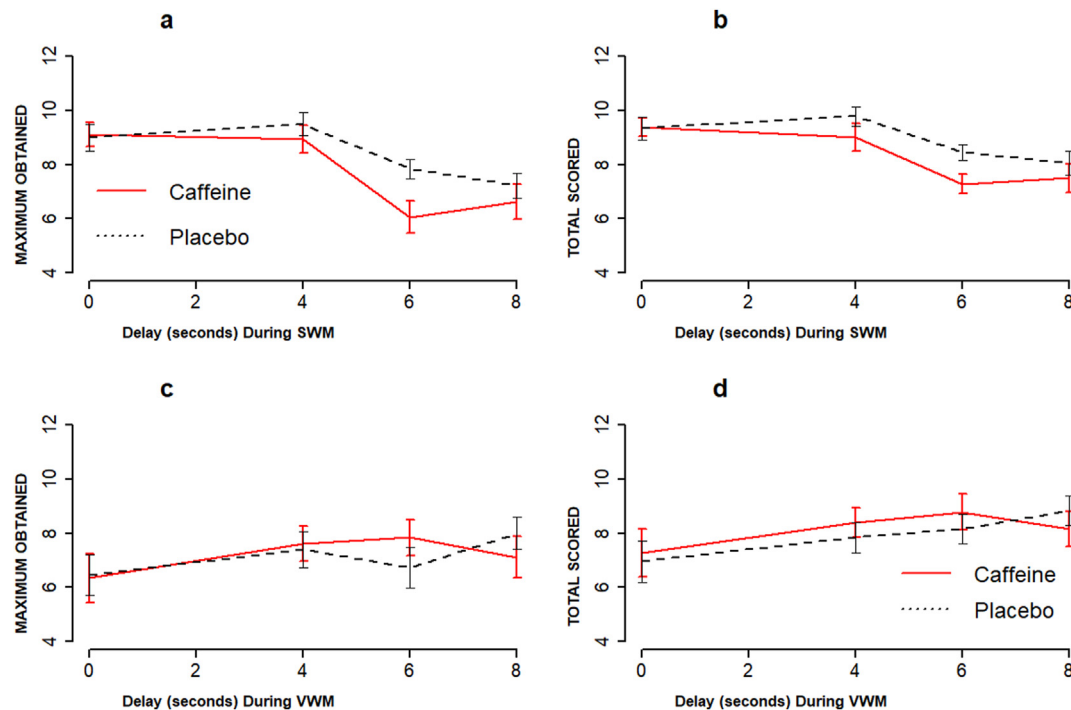


Fig. 1. The effects of caffeine (200 mg, PO) on spatial working memory (SWM, a and b) and verbal working memory (VWM, c and d) during span backward tasks. Data are presented as mean $\pm$ standard error of mean (SEM, $n = 18$) while the confidence interval was 95%. Maximum obtained for SWM (a), total scored for SWM (b), maximum obtained for VWM (c), and total scored for VWM (d). For SWM, there were significant main effects of caffeine, relative to the placebo ($p < 0.05$, for maximum obtained and total scored). There were also significant effects of caffeine on maximum obtained ($p = 0.01$) and total scored ($p = 0.06$) at 6 s delay.

[1.6, 26] = 3.6, $p = 0.02$, $\eta^2 = 0.05$), but failed to be significant for the maximum obtained ($F [3,51] = 2.5$, $p = 0.07$, $\eta^2 = 0.028$). There were also no significant drug-by-delay interactions for maximum obtained ($F [3,51] = 1.6$, $p = 0.20$, $\eta^2 = 0.03$) and total scored ($F [3,51] = 1.2$, $p = 0.31$, $\eta^2 = 0.02$). The linear mixed-effects model (with Satterthwaite's method) showed near significant effects of session on the total scored ($F [1,16] = 3.6$, $p = 0.08$, $d = 0.07$) and maximum obtained ($F [1,16] = 3.6$, $p = 0.07$, $d = 0.04$) for VWM.

4. Discussion

The current investigation indicates that consuming caffeine-containing beverages on a regular basis could potentially negatively impact SWM performance. A deleterious effect of caffeine on SWM was observed during the backward task, suggesting that it may have unfavorable consequences on information manipulation but not on information encoding or maintenance. Regarding the negative WM effects of caffeine in habitual caffeine consumers, our finding is comparable with a recent study that found that caffeine impaired WM performance in habitual coffee drinkers in n-back tasks [16]. Although the findings, participants' age range, and sample size were comparable, it has to be noted that there are important methodological differences between Lin and colleagues' study [16] and the present study. First, they have used n-back tasks which commonly measure visual WM, while we used SST. Second, they have recruited participants with higher (~475 mg) daily caffeine consumption history than participants in the current study (1.6 cup of coffee or ~100–200 mg caffeine). However, it can be comprehend that the total administered caffeine doses (450 mg and 200 mg) in each study, respectively, was proportional to the participants daily caffeine intake. Lastly, they have administered caffeine for ten consecutive days, while we administered a single dose of caffeine. Despite methodological and task differences, the

results indicate that caffeine administration in regular coffee/tea consumers might have deleterious effects on WM performances. However, most previous studies on the acute effects of caffeine on VWM using n-back or forward DST have reported no significant effects of caffeine on WM [11,17,30–33]. The absence of caffeine effect on VWM manipulation using backward DST might be linked to the lesser sensitivity of drugs to digit tasks compared to spatial tasks for measuring WM [23,34,35]. In contrast, another study [36] found that acute administration of low-dose canned coffee, containing approximately 67 mg of caffeine, in 21 healthy participants enhanced VWM on the forward and backward DST and executive function on the Trail Making Test. It should be noted that this study was not a double-blind, placebo-controlled trial and the caffeine consumption history of the participants was not mentioned [36]. The discrepancies observed in studies investigating the consequences of caffeine intake on cognitive performances can be attributed to caffeine's differential impact on both lower-and higher-cognitive functions, such as accuracy in attention tasks [2,37–41] or WM [16], respectively. The present finding also supports a previous suggestion that caffeine might impair performances that highly depend on WM [38].

The current finding also adds an important implication for the growing concern on the impact of adenosine and caffeine, given the role of adenosine in brain functions (and dysfunctions) and the action of caffeine on adenosine receptors (ARs) [42–47]. Caffeine is known to non-selectively antagonize ARs [48,49] which in turn counteracts the effects of adenosine on firing neurons [50–52]. While it is too early to draw a definitive conclusion about the relationship between caffeine and the “adenosine hypothesis of schizophrenia” (for review, see [53]) and precaution is needed, dysfunction of the purinergic system can lead to a decrease in adenosinergic activity, which may result in similar impairments observed in individuals with psychosis-like symptoms [43]. For

instance, some evidence have suggested that an antagonistic interaction between ARs and dopaminergic receptors in the pre-frontal cortex or striatum could negatively impact brain functions [54–57]. Furthermore, it is important to mention that caffeine can potentially affect WM indirectly by altering cortical or subcortical activations during WM tasks, an area that requires further investigation [16,31].

The limitations and strengths of this study should be noted. The main strength of the present study was its study design and the use of the delay condition to examine WM. However, there are important limitations. One, the relatively small sample size may have affected the reliability of the study. Two, it has to be noted that most of the participants were young people that limit the generalisability of the results. Three, the number of male and female participants is not balanced in the present study that might affect the results observed, although some previous studies indicated that cognitive or physiological effects of caffeine is sex-independent [58–60]. Furthermore, to determine whether the effect of caffeine was due to attention or WM, it will be crucial in future studies to administer DST and SST in healthy participants while presenting interference. Additionally, using longer delay conditions may reveal delay and drug effects or their interactions.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declare that there is no conflict of interest.

Acknowledgments

The University of Western Australia (RTP-HDR) supported this study during data collection and the author received stipend and other resources from Linköping University during retrospective conceptual design, data analysis, manuscript write-up, and submission and revision periods. The author appreciates Drs Jennifer Rodger, Matthew Albrecht and Mathew Martin-Iverson for supervising the project during original data collection. The author also would like to thank each volunteer for their participation.

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